

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech. Branch: Civil / Civil & Infra. / Civil & Environmental Engineering

Subject Code & Name: BTCEC305 / BTCVC305 / BTCIC305 Surveying

Max Marks: 60

Date: 09/07/2024

Semester: III

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define surveying. Write down applications of surveying. State the principles of Surveying.	CO1	6
B) What is meant by Ranging? Explain the methods of ranging in detail.	CO1	6
C) Explain in detail Errors in Chaining.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Define the following terms: i) True Meridian ii) Magnetic Bearing iii) Magnetic Meridian iv) True Bearing v) Arbitrary Meridian vi) Local Attraction	CO2	6
B) Convert the following WCB's to QB's a. WCB of AB = $45^{\circ}30'$ b. WCB of CD = $108^{\circ}45'$ c. WCB of GH = $75^{\circ}50'$ d. WCB of HK = $145^{\circ}20'$ e. WCB of ML = $340^{\circ}10'$ f. WCB of DE = $60^{\circ}10'$	CO2	6
C) Enlist the different methods of Plane Table and Explain any two in detail.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Write down the characteristics of contour lines.	CO3	6
B) During a levelling work started from a BM of known R.L.=100m. The following staff readings were obtained: 0.850, 1.555, 1.725, 0.455, 1.800, 1.750, 0.950, 1.555 The instrument was shifted after 3rd and 6th readings. Enter correctly all the readings in a field book page and determine the reduced levels of all stations showing calculations by Rise and Fall method. Apply usual arithmetic check.	CO3	6

C) What are the types of levelling operations explain with neat sketch. CO3 6

Q.4 Solve Any Two of the following. 12

A) What are the different methods of traversing Explain any one in detail. CO4 6

B) Explain Bowditch Rule and Transit Rule. CO4 6

C) Define (any three): CO4 6

1. Transiting of Theodolite 2. Swinging 3. Line of Collimation
4. Telescope Normal 5. Face left 6. Face right

Q. 5 Solve Any Two of the following. 12

A) Explain the Steps involved in survey of Waterways. CO4 6

B) Explain the steps involved while carrying out Engineering Survey. CO4 6

C) Write a short note on Reconnaissance. CO1 6

***** End *****